Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	September 2026
Project Name	ServiceNow-Based Employee Laptop Request Application

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2

Project: ServiceNow-Based Employee Laptop Request Application

Reference: https://www.servicenow.com/

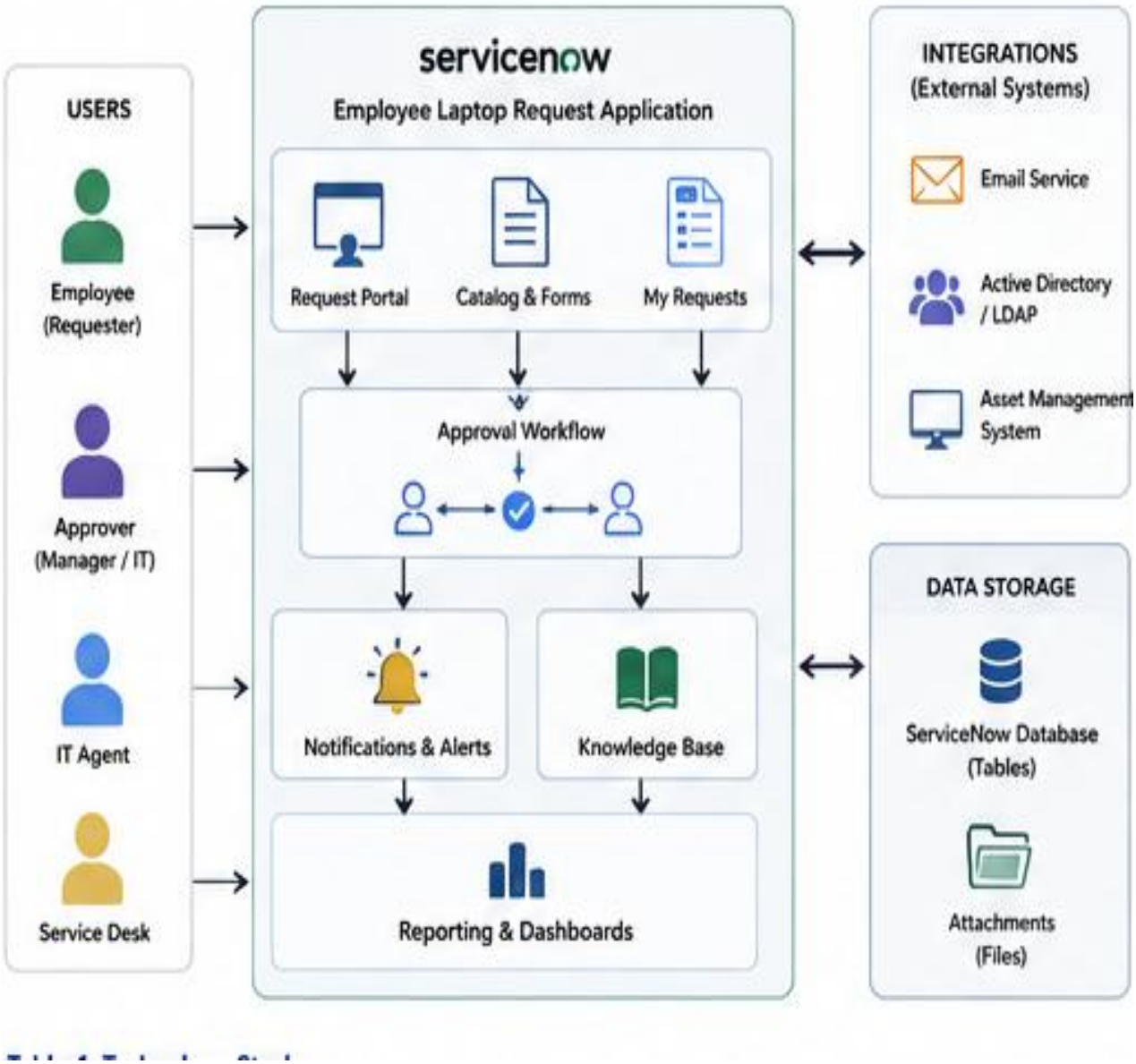


Table-1: Technology Stack:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application (e.g., Request Portal, Catalog, Agent Workspace, Mobile UI, Notifications).	ServiceNow UI (Now Experience), Service Portal, Responsive UI, Mobile App (Now Mobile)
2.	Application Logic - 1	Business logic for raising laptop request, validations, and workflow initiation.	ServiceNow Flow Designer, Business Rules, Script Includes (JavaScript)
3.	Application Logic - 2	Approval workflow logic (manager, IT approvals) and task orchestration.	ServiceNow Workflow, Flow Designer, Approval Policies
4.	Application Logic - 3	Notification logic for updates and reminders.	ServiceNow Notifications, Event Management
5.	Database	Application data storage (requests, tasks, approvals, assets, users, etc.).	ServiceNow Database (Tables)
6.	Cloud Database	Platform database service in cloud.	ServiceNow Cloud (Oracle DB underlying)

S.No	Component	Description	Technology
7.	File Storage	Storage for attachments like justifications, invoices, documents.	ServiceNow Attachment Storage (OCI Object Storage)
8.	External API - 1	Used to send email notifications and reminders.	SendGrid / Microsoft Graph API / SMTP API
9.	External API - 2	Used for user data sync / validation with corporate directory.	Microsoft Graph API / Active Directory (LDAP) API
10.	Machine Learning Model	Used for request categorization / suggestions (if applicable).	ServiceNow Predictive Intelligence (Optional)
11.	Infrastructure (Server / Cloud)	Application hosted on ServiceNow cloud platform.	ServiceNow Cloud (Multi-tenant) on AWS

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used.	N/A – ServiceNow is a proprietary platform. Uses open-source components internally (e.g., Angular, Bootstrap).
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	Role-Based Access Control (RBAC), ACLs, Data Policies, Encrypted Communication (HTTPS), SSO (SAML), MFA, IP Restrictions
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	ServiceNow Multi-tenant Cloud Architecture, Horizontal Scalability, Load Balanced Infrastructure
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	ServiceNow High Availability, 99.9% Uptime SLA, Load Balanced Instances, Automated Failover
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's etc.).	ServiceNow Platform Performance Analytics, Caching Mechanism, CDN for static content, Optimization of Business Rules & Queries



References:

- https://www.servicenow.com/
- https://docs.servicenow.com/
- https://developer.servicenow.com/
- https://www.servicenow.com/solutions/employee-experience.html
- https://www.servicenow.com/products/it-service-management.html
- https://www.servicenow.com/solutions/asset-management.html
- https://www.servicenow.com/solutions/workflow-automation.html
- https://www.servicenow.com/solutions/security-operations.html

Project Technology Scope Note:

This project will be built entirely on the ServiceNow Platform, leveraging its ITSM, Workflow Automation, Catalog Management, and Integration capabilities to deliver a seamless and secure Employee Laptop Request experience. All data will be stored in ServiceNow cloud infrastructure with role-based access, automated workflows, and real-time notifications.

Key Modules / Capabilities Covered:

